

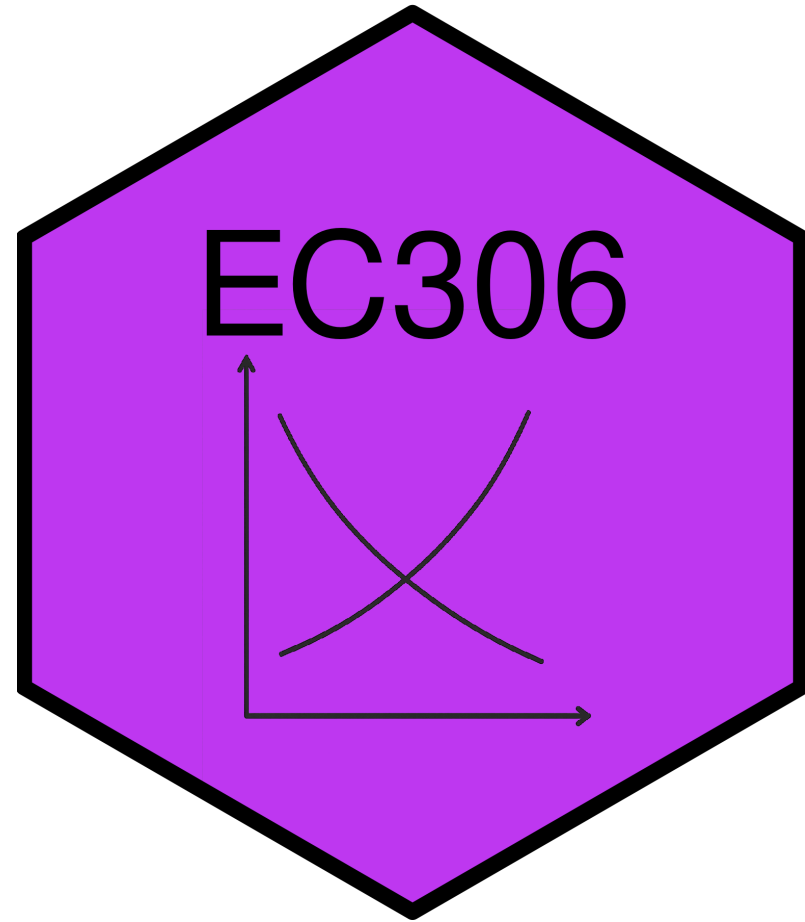
Human Capital Theory

EC306- Wages and Employment

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Fall 2025



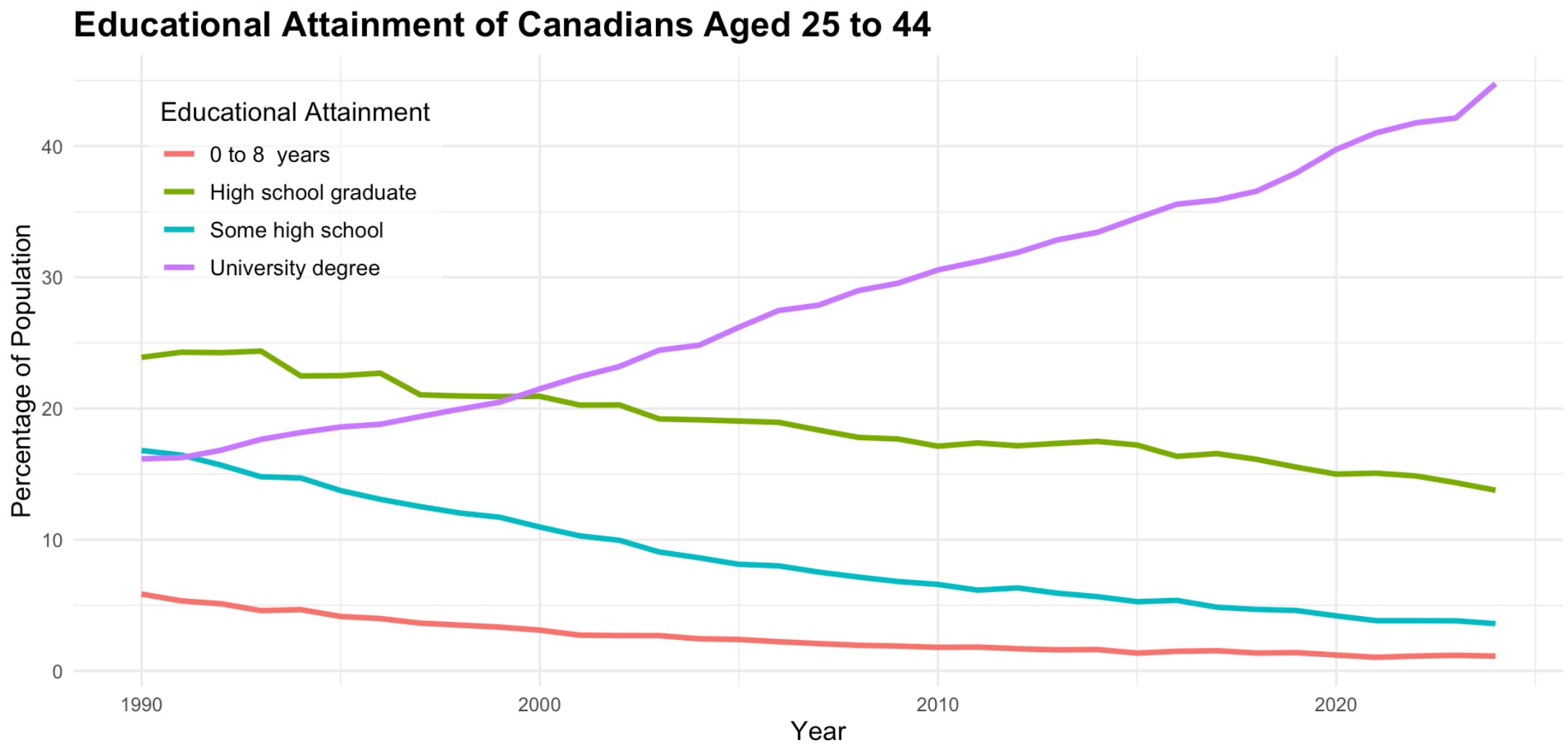
Goals of This Section

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- Bring together demand and supply to determine wages and employment
- Discuss the effect of market power in goods market on equilibrium wages and employment
- Discuss the effect of market power in the labour market on equilibrium wages and employment
- Analyze minimum wages

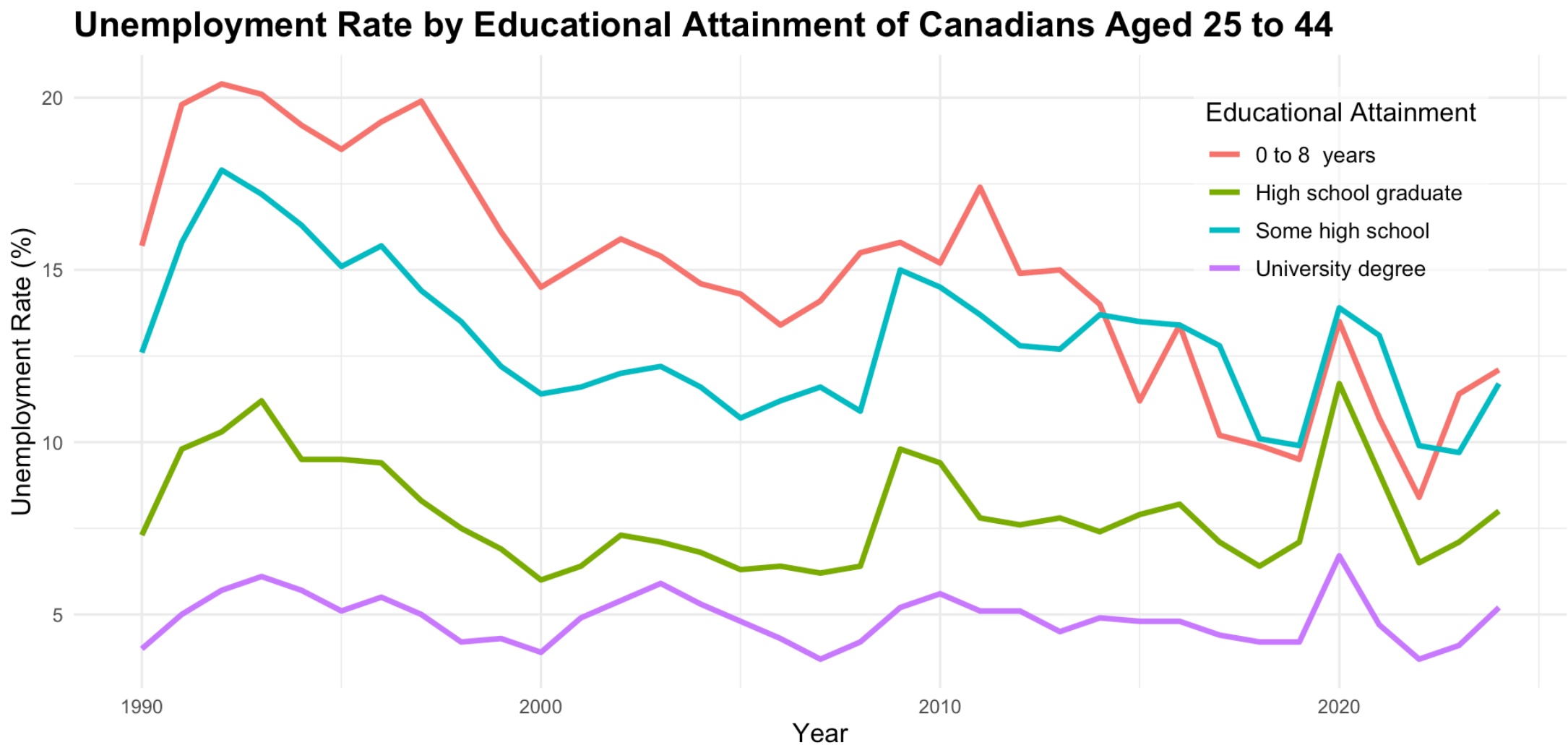
Introduction

Educational Attainment



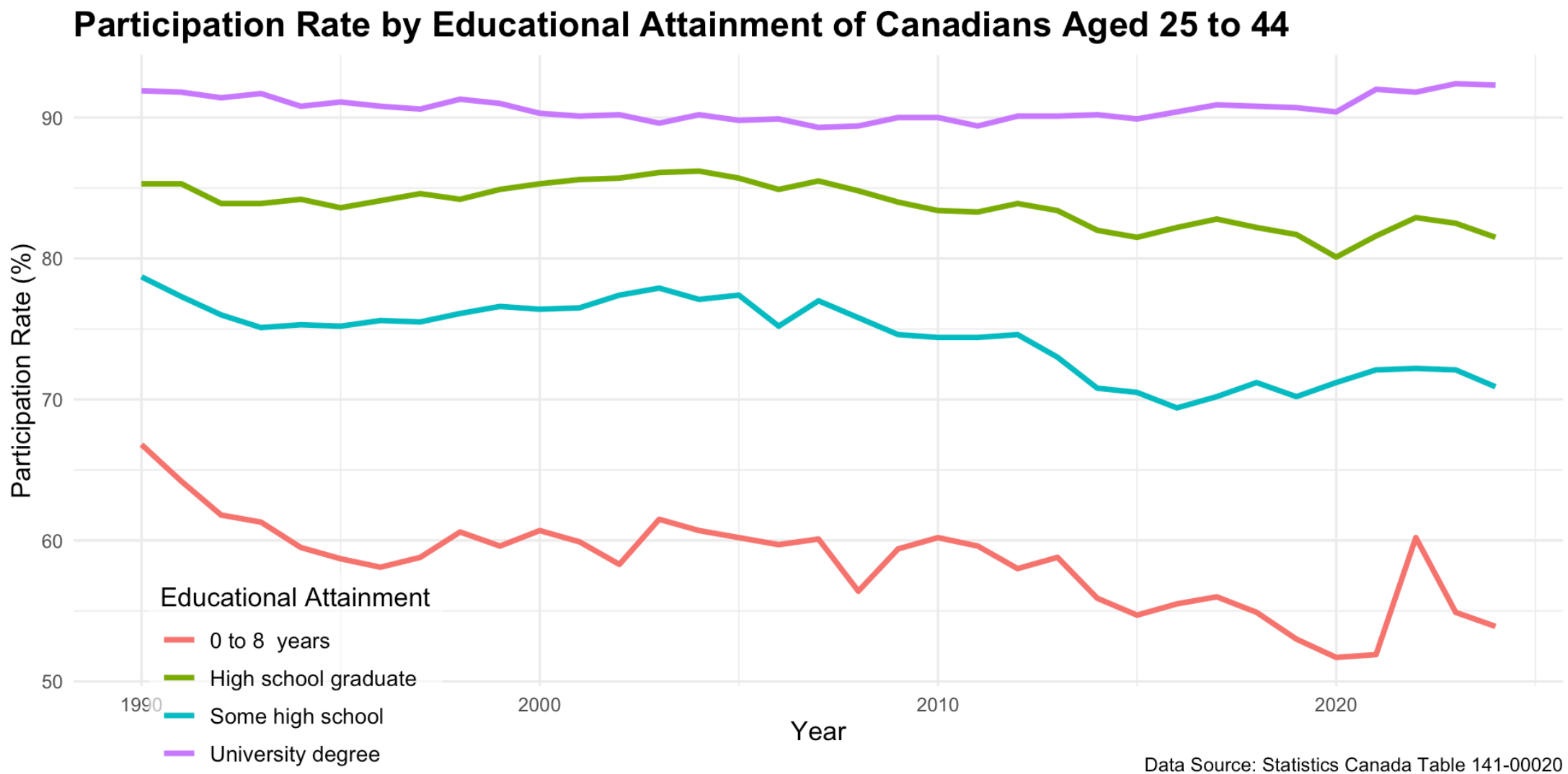
Data Source: Statistics Canada Table 141-00020

Unemployment Rate



Data Source: Statistics Canada Table 141-00020

Participation Rate



Human Capital Theory

Human Capital Theory

- Firms often invest in physical capital (e.g., machines, tools, buildings)
 - Incur some cost today to invest in capital
 - Boosts the productive ability of workers
 - Increases the output flow in the future
- Decision to invest is made by comparing costs to benefits
 - Invest if benefits outweigh costs
 - Costs are usually upfront, while benefits occur slowly over time

Human Capital Theory

- Individuals make similar investments in *human* capital
 - **Human Capital:** An individual's stock of productive skills, knowledge, and abilities
 - People incur upfront costs for education, training, etc.
 - Training improves productivity, and therefore income
 - Individuals earn higher income streams in the future
- We will study investments in human capital in a similar way to physical capital

Human Capital Theory

- Human capital investments involve some special considerations
 - Non-wage benefits
 - Consumption value
 - Differences in risk of unemployment in downturns
 - It is more difficult to finance education investments
 - No collateral
- Social versus private costs/benefits
 - Private costs and benefits are specific to the individual
 - But there may be social costs and benefits as well
 - E.g., positive externalities from education

Private Investments In Education

Introduction

- The human capital investment decision boils down to comparing lifetime income streams
 - Different streams of income are associated with different levels of schooling
 - Higher levels of schooling usually lead to higher income streams
- We will outline a process to decide which stream to choose

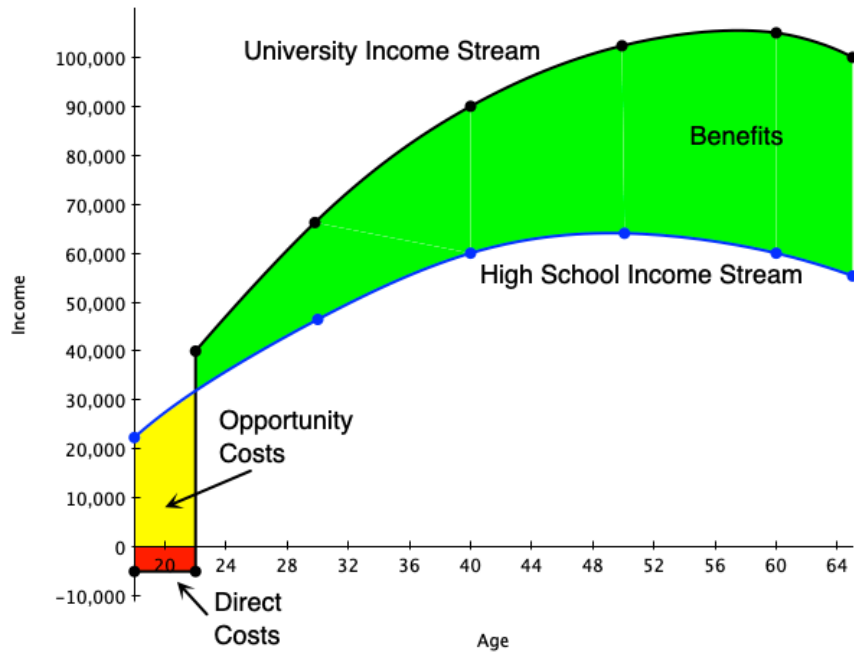
To make the model tractable, we assume:

1. No direct utility or disutility from schooling
2. Hours of work are held fixed
3. Certainty of income streams
4. Individuals can borrow and lend at interest rate r

Attending University

- Suppose we want to analyze whether it is financially worthwhile to attend university
 - Start from the perspective of an 18-year-old high school graduate
 - This student is deciding whether or not to attend university
 - Students who do not attend university move immediately into the workforce
- The graph on next slide depicts two income streams:
 - **Black line:** university path
 - **Blue line:** high school path
 - **Red and yellow:** costs of attending university
 - **Green:** benefits of university education

Attending University



- Direct costs are in red
 - Books, fees, tuition
- Foregone wages are in yellow
 - Income not earned during the time in university
- Net gains in wages are in green
 - Earnings of university grad above a high school grad
- Beneficial to attend if present value of green exceeds present value of yellow plus red

Attending University

- People will choose the level of schooling that maximizes the net present value of lifetime earnings
- We illustrated that graphically above
- Mathematically, express the present value of earnings for a high school graduate as

$$PV = \frac{Y}{(1+r)^0} + \frac{Y}{(1+r)^1} + \dots + \frac{Y}{(1+r)^{T-18}}$$

- In this equation
 - Y = annual earnings of high school graduate
 - r = interest rate
 - T = expected retirement age

Attending University

- We can approximate the present value of earnings for a high school graduate as

$$PV \approx Y + \frac{Y}{r}$$

- Approximation works if T is long enough
- Now consider a person who attends one year of university
- They incur direct costs D in the first year and earn no income in that year
- Every year they work, they gain an amount ΔY above the high school graduate

Attending University

- Their income stream is

$$PV^* = \frac{-D}{(1+r)^0} + \frac{Y + \Delta Y}{(1+r)^1} + \dots + \frac{Y + \Delta Y}{(1+r)^{T-18}}$$

- This can be approximated as

$$PV^* \approx -D + \frac{Y + \Delta Y}{r}$$

- The net present value of attending this additional year of university is

$$NPV = PV^* - PV \approx -(D + Y) + \frac{\Delta Y}{r}$$

Attending University

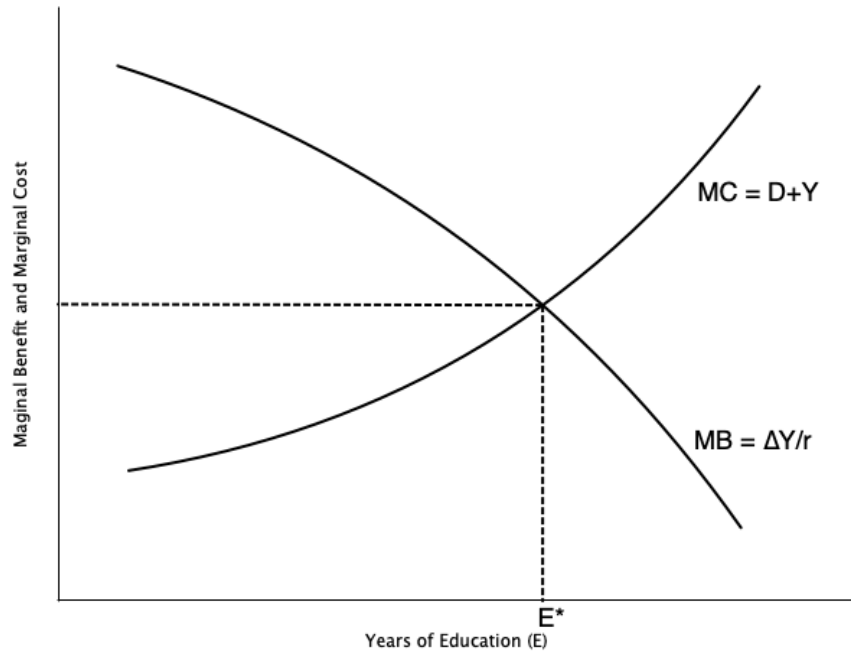
- The term $D + Y$ is the marginal cost of attending university for one year
- The term $\frac{\Delta Y}{r}$ is the marginal benefit of attending university for one year
- Attending university for one year is worthwhile if

$$NPV > 0 \implies \frac{\Delta Y}{r} > D + Y$$

- A person will continue to attend more years of schooling until $MB = MC$, or

$$\frac{\Delta Y}{r} = D + Y$$

Attending University



- Graph on left illustrates optimal schooling decision
- Marginal benefit curve is downward sloping
 - Each additional year of schooling yields a smaller increase in wages
- Marginal cost curve is upward sloping
 - Each additional year of schooling incurs higher direct costs, foregone wages, psychic costs
- Optimum is where $MB = MC$

Attending University

- Another way to look at the schooling decision is through the rate of return to schooling
- This method uses the concept of the internal rate of return (IRR)
- The IRR is the interest rate that makes the NPV of an investment equal to zero
- In our setup, the IRR is the interest rate that satisfies

$$0 = -(D + Y) + \frac{\Delta Y}{IRR}$$

- Rearranging gives

$$IRR = \frac{\Delta Y}{D + Y}$$

- The IRR is the percentage return on the investment in schooling

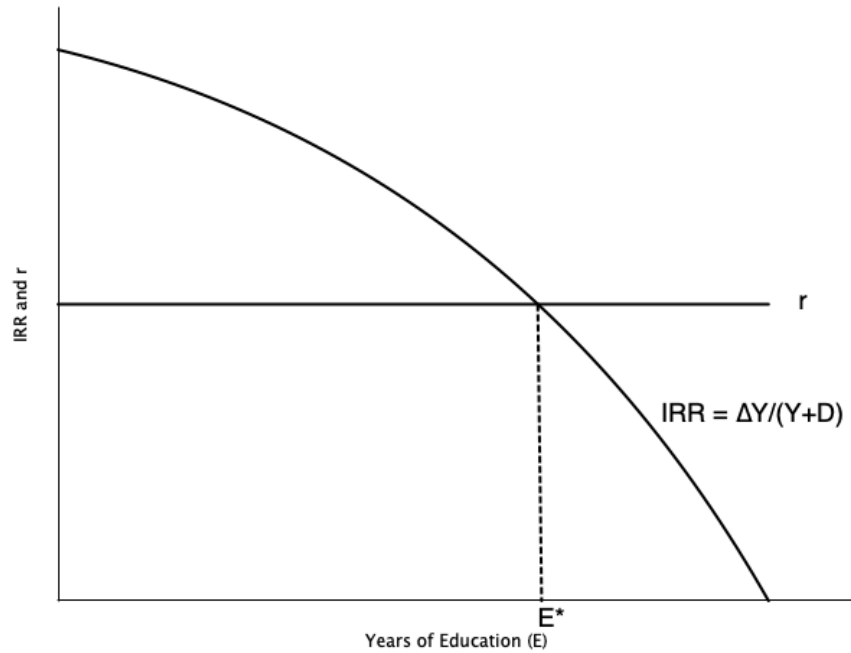


Attending University

- The alternative to attending university is to invest the money $D + Y$ at the market interest rate r
 - Opportunity cost of investing in schooling is r
- If the IRR exceeds r , then investing in schooling is worthwhile
- A person would continue to attend more years of schooling until

$$IRR = r \implies \frac{\Delta Y}{D + Y} = r$$

Attending University



- Graph on left illustrates alternative optimal schooling decision
- Market interest rate is r
- Internal rate of return IRR is decreasing in schooling
 - Each additional year of schooling yields a smaller increase in wages
- Optimum is where $IRR = r$
- Yields same optimal schooling level as previous method

Comparative Statics

- We can use this framework to see how changes in parameters and policies affect schooling decisions
- Suppose the direct costs D of attending university decrease
 - E.g., government introduces tuition subsidies
 - This decreases the marginal cost of schooling
 - Leads to an increase in optimal schooling level
- Suppose the market interest rate r decreases
 - E.g., due to monetary policy
 - This increases the present value of future earnings
 - Leads to an increase in optimal schooling level

Comparative Statics

- What if the income tax system becomes more progressive?
 - Higher taxes on high income earners
 - Decreases the after-tax increase in earnings ΔY
 - Leads to a decrease in optimal schooling level
- Suppose the pay for jobs in the oil sector increases substantially
 - E.g., due to a boom in oil prices
 - Increases the earnings of high school graduates Y
 - Leads to a decrease in optimal schooling level

Complications

- Information asymmetries
 - Students may not know the true costs and benefits of schooling
 - May lead to under- or over-investment in schooling
- Disutility of schooling
 - Students may dislike attending school
 - This adds to the marginal cost of schooling
- Different degrees have different returns
 - Some fields of study lead to higher earnings than others
 - Students must choose not only how much schooling, but what type

Complications

- A major complication is financing/credit constraints
 - Many students cannot afford the upfront costs of schooling
 - May not be able to borrow against future earnings
 - Leads to under-investment in schooling compared to optimal level

Signalling

Introduction

- In the Human Capital model, there are key assumptions that make the model work:
 - Education increases productivity
 - Productivity is perfectly observed by employers
 - Wages are paid according to productivity
- The **signalling model** emphasizes an alternative view:
 - Education may not increase productivity
 - Instead, it reveals productivity to employers
 - It acts as a signal of worker ability
 - People only invest in education to signal their ability to employers

Model

- The basics of the model are
 - Employers do not observe productivity until after a worker is hired
 - Employees know their own ability but find it difficult to credibly reveal it to employers
 - Without knowing productivity, firms pay employees according to *average* productivity in the population
 - Under certain conditions, obtaining education can serve as a *signal* of productivity, allowing workers to be paid accordingly

Model

- For education to actually be informative about ability:
 - It must be costly to acquire
 - It must be **sufficiently more costly** for low-ability individuals than for high-ability individuals
 - Otherwise, low-ability workers might find it profitable to *fake* high ability by acquiring education
 - Employers' beliefs must be confirmed in the future
 - Otherwise, they will update their wage offers
- Given the conditions above, education helps solve an information problem:
 - It correctly reveals ability through educational attainment
 - High- and low-ability workers are then correctly paid according to their productivity

Model

- Below we do a more formal treatment of the model
- Assume that:
 - There are two types of workers: low ability (L) and high ability (H)
 - A fraction q are Type L, and $(1 - q)$ are Type H
 - For Type L workers:
 - Cost of education: $c_L(s) = s$
 - Marginal Product (MP): 1
 - For Type H workers:
 - Cost of education: $c_H(s) = \frac{s}{2}$
 - Marginal Product (MP): 2

Model

- Employees know their own ability, but employers do not
- Employers instead form *beliefs* about employee ability based on observed education levels
- What would happen if there were no way to signal ability?
- Firms would have no way of knowing who is high or low ability
 - So they offer everyone the same wage.
- The wage equals the **average productivity**:

$$\bar{w} = 1 \cdot q + 2 \cdot (1 - q) = 2 - q$$

Model

- Example: 50% Low-Ability Workers
 - If 50% of the population are low ability ($q = 0.5$):

$$\bar{w} = 2 - q = 1.5$$

- Recall that
 - Type L workers produce $MP = 1$
 - Type H workers produce $MP = 2$
- Therefore:
 - Type L is paid **more** than they produce
 - Type H is paid **less** than they produce
 - On average, total pay equals total production

Model

- Suppose employers observe education
- They form beliefs about how it relates to ability, and pay according to those beliefs.
- One possible set of employer beliefs:
 - Workers with education $s > s^*$ are Type H
 - Workers with education $s \leq s^*$ are Type L
 - s^* is some cutoff level of education
- Firms pay wages according to the believed type:

$$w = \begin{cases} 2, & \text{if } s > s^* \\ 1, & \text{if } s \leq s^* \end{cases}$$

Model

- In this model, workers decide how much education to acquire.
- A worker of type (t) maximizes utility:

$$U_t(s) = w(s) - c_t(s)$$

- All workers can obtain $U_t(s) = 1$ by acquiring **no education**.
 - Therefore, the payoff to acquiring any education must satisfy $U_t(s) \geq 1$
- Workers will never find it useful to obtain **more than** s^* years of education.
 - Doing so increases costs without increasing payoffs.
- Thus, the decision is binary: either **0** or s^* years of education.

Model

- Type H Workers

$$U_H(0) = w(0) - c_H(0) = 1 - 0 = 1$$

$$U_H(s^*) = w(s^*) - c_H(s^*) = 2 - \frac{s^*}{2}$$

- A Type H worker will acquire s^* years of education if:

$$U_H(s^*) > U_H(0)$$

$$2 - \frac{s^*}{2} > 1$$

$$s^* < 2$$

Model

- For Type L Workers

$$U_L(0) = w(0) - c_L(0) = 1 - 0 = 1$$

$$U_L(s^*) = w(s^*) - c_L(s^*) = 2 - s^*$$

- A Type L worker will acquire s^* years of education if:

$$U_L(s^*) > U_L(0)$$

$$2 - s^* > 1$$

$$s^* < 1$$

Model

- If $s^* < 1$:
 - Both Type L and Type H acquire education s^*
 - The signal is **not informative**
- If $s^* > 2$:
 - Both Type L and Type H acquire **no education**
 - The signal is **not informative**
- If $1 < s^* < 2$:
 - Type H obtains education s^*
 - Type L obtains no education
 - Education **reveals worker type**

Model

- Equilibrium
- Occurs when s^* lies **between 1 and 2**.
 - This is an equilibrium because the beliefs based on this strategy are **correct**
 - A firm can only maintain beliefs if they are **validated** by observed behaviour
- Any s^* between **1 and 2** will satisfy equilibrium conditions
 - The equilibrium with the **highest welfare** is the one closer to $s^* = 1$
- This is called a **separating equilibrium**.
 - When everyone gets the same level of education, it is a **pooling equilibrium**.
 - The case where there is **no education signal** is one example of a pooling equilibrium.

Model

